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Assessment Task – Portfolio (Learner Version)

Assessment Task 2: Securing Cloud Services

There are THREE (3) parts in this assessment. You should complete all tasks.

- Part 1: Security Implications based on Deployment and Delivery Models
- Part 2: Factors to consider when assessing a Cloud Service provider.
- Part 3: Create, Assess and Secure Cloud Services

Part 1: Security Implications based on Deployment and Delivery Models

To understand the level of vulnerability in a cloud computing platform, it is critical to consider two key factors: the choice of **deployment** and **delivery** model. As you may already know, the industry recognises three standard deployment models and three delivery models. Each of these models has distinct security implications. Your objective is to gather relevant information and engage in a discussion about each deployment and delivery model, covering their **description**, **implications**, and **challenges**. Please complete the corresponding sections in **Table A** and **Table B** provided below.

Table A Cloud Deployment Models

Deployment Type	**Description**	**Implications**	**Challenges**
Private Cloud	Your response: It is a cloud computing which let one organization only to operate on the cloud infrastructure without sharing its resources with others. It can be managed by the organization or by a third-party and can be hosted on-premises or in a third-party datacentre.	Your response: - Higher cost due to the need to buy the hardware, software and infrastructure. - Due to the control and benefits it offers, the security of the information could be managed better. - Gives a better performance as the resources are not shared.	Your response: - Higher cost for buying the components and maintaining them. - Require an IT team for managing the private cloud. - Limitations of scalability as it can be expensive to purchase new hardware. - The private cloud might won integrate well with some existing systems.
Public Cloud	Your response: It is a cloud computing type which is providing services over the internet and share them across. The services are managed by a third-party cloud provider which are managing the infrastructure. The services can be accessed from the web browser and the services can cost based on pay-as-you-go.	Your response: - Giving a better cost management as every service got pay-as-you-go option which can be managed according to the usage of the organization. - Gives better flexibility as changes can be adapted quickly. - Gives great scalability options based on real-time demand.	Your response: - Due to the data be over the internet, it can be more vulnerable to data breaches or unauthorized access. - The cost of pay-as-you-go pricing can be hard to predict and can lead to unexpected high cost. - Performance can be affected by many factors such as internet performance. - Downtime service of the provider could affect business operations. - Can be challenging to change providers which

			can risk the organization with vendor lock-in.
Hybrid Cloud	Your response: It is a combination of the private cloud and the public cloud. It ensures the customer can enjoy the benefits of both deployments for best scalability and flexibility. The data can be moved and shared between the private and the public cloud.	Your response: - Due to the multiple environments, it offers a good level of disaster recovery. - Better performance due to running workloads in the right environment. - Be able to have the benefits of both deployments.	Your response: - Can be complex to manage due to handling multiple environments. - Security maintenance can be complex across both deployments. Security policies, tools and best practices are required. - Cost could be higher due to the management of resources across environments. - Performance can be affected by many factors such as internet performance. - Can be challenging to change providers which can risk the organization with vendor lock-in.

Table B Cloud Delivery Models

Delivery Type	Description	Risk and Challenges
Infrastructure as a Service (IaaS)	Your response: It is a service model in cloud computing which provides virtualized computing resources over the internet such as VMs.	Your response: The data is stored over the internet on the cloud service which can lead for cyberattacks if security measures are not implemented well. If the user won't understand the shared responsibility model, it can lead for misunderstanding and security gaps. The data could be compromised by an insider threat from the provider or customer employees. Weak permissions and misconfigurations configured by the customer could also lead for vulnerabilities. Also, the customer's operations could be interrupted by the service provided due to downtime or outages.
Platform as a Service (PaaS)	Your response: It is a cloud computing service model which gives the customer the ability of developing, running and managing applications without the need to deal and maintain complex underlying infrastructure.	Your response: data security can be a challenge while the data is moving between the user's environment and the PaaS provider. Unauthorized access to the production environment which is stored on the cloud could lead to breaches. Also, misunderstanding of the shared responsibility model between the customer and the providers for the securing of the applications being build on the platform, could leaf security gaps and cyberattacks. Also, the customer's operations could be interrupted by the service provided due to downtime or outages.
Software as a Service (SaaS)	Your response: It is a cloud computing service model which give the ability to get applications available over the internet via web browser.	Your response: Data privacy could be breached if not following data protection regulations. Also, sensitive data could be stored in the cloud and can be exposed if strong security measures are not taken into consideration. Vulnerabilities in the application or the underlying infrastructure could be exploited by attackers. Also, an insider threat from the service provider or the customer's organization could be a risk for the security of the application. In addition, the

		service will be dependent on the provider reliability and performance, which could lead to business operations disruptions.

Part 2: Factors to consider when assessing a Cloud Service provider.
Read the case study below and answer the questions that follow. Your task is to identify the Australian Act that has been violated in this incident and consider other factors that should be taken into account.

Case Study - Australian healthcare service data breach
In 2016, a company providing cloud services to a national healthcare service inadvertently exposed the details of 550,000 people online. A backup of a database containing personal information was accidentally saved to a public-facing web server. The cloud service provider was found to be in data breach by disclosing personal information and failing to take reasonable steps to protect personal information.

The Office of the Australian Information Commissioner (OAIC) acknowledged the response by both parties to this incident was prompt and well managed. While the cloud service provider was responsible for failing to protect personal information, the healthcare service was also found to be in breach of the Privacy Act. This was due to not taking reasonable steps to ensure adequate security measures were taken by the cloud service provider, and for having a data retention period that was longer than necessary. All affected individuals were contacted, information handling practices were improved, and information security was strengthened to mitigate against future breaches of this nature.

Question 2.1:	Identify the Australian Act that has been violated in this incident.
Your response:	The Australian Privacy Act 1988.
Question 2.2:	Which specific principle within the Act has been breached?
Your response:	- APP 1: open and transparent management of personal information. - APP 6: Use or disclosure of personal information. - APP 11: Security of personal information. - APP 12: Access to personal information. - APP 13: Correction of personal information. Resources: https://www.oaic.gov.au/privacy/australian-privacy-principles/read-the-australian-privacy-principles
Question 2.3:	Explain why the cloud service provider was found to be in breach of the Privacy Act.
Your response:	Because the cloud service was failing to take the right steps to protect personal information which led in this data breach to the disclosure of personal information with 550,000 personal details being exposed. The backup data with the personal information was saved by mistake on a public web server.
Question 2.4:	Discuss the reasons why the healthcare service was also found to be in breach of the Privacy Act.
Your response:	The healthcare service failed to make sure that the service provider is complying with the data protection requirements such as security measures before choosing this provider. According to the Privacy Act, the organizations are responsible to make sure their providing take the right steps when handling personal information. Also, the healthcare service kept sensitive data stored for longer than the data retention period which was necessary.
Question 2.5:	What were the consequences of the breach for the affected individuals?
Your response:	Individuals could face a variety of consequences such as personal information exposure which might include names, addresses and sensitive health information along with a potential identity theft depending on the data which was on the system. Also, their details could be used to gain access to financial accounts such as bank accounts which could lead to financial fraud.
Question 2.6:	Outline the steps taken by both parties to address the incident.
Your response:	The healthcare provided contacted all affected individual and informed them about the data breach, about which of their details got exposed and which steps are being taken and needs to be taken by them to minimize

the impact. Also, the organization reported the incident to the OAIC and worked with them together to manage this breach and comply with all legal requirements. Furthermore, they revied their policies and practices and update them accordingly to make sure that the lesson is learned including data retention period. They also managed the incident with the cloud provider.

The cloud service provider likely to use an incident response plan to minimize the impact of the breach, investigate its internal procedures, communicate with the healthcare provided to share findings and steps which were taken, strengthen the security with better encryption for instance and train its stuff for avoiding future incidents.

Question 2.7:	What measures could have been implemented by the healthcare service to prevent this breach?

Your response:

- Assessing multiple cloud services available to understand better their security practices and to ensure they can handle sensitive data securely. This will ensure that the provider that the company choose is the best option. Also, reviewing the cloud service provider regularly to make sure their ongoing compliance with security requirements.
- Encrypting data at rest and in transit and use encryptions for all backups to avoid storing backups on public-facing servers which happened before.
- Apply least privileges and ensure strict access control along with MFA enabled and the monitoring of the access by reviewing logs for suspicious activities.
- Making sure do not keep personal data which is not required for operational purposes and removing any data which is not necessary by implementing regular data purging.
- Have a comprehensive incident response plan for identifying risks in time, containing them and mitigating along with the right training and awareness programs for employees.
- Reviewing and updating policies and procedures regularly.
- Research and implement advanced security technologies to minimize the risk of cyber incidents.

Question 2.8:	Why is it important for organisations to comply with privacy laws and protect personal information?

Your response:

- Complying with privacy laws and protecting personal information can avoid legal consequences in case of incident which happened due to the non-compliance with the law. It can lead to fines, penalties, lawsuits and reputational damage.
- Helping with the respect of privacy toward their customers and the prevention of any harm due to a data breach.
- Making sure the customer trusting the organization and to ensure the ongoing trust.
- Protecting the business reputation, operations and financial sate which could be damaged by cyber incident because not complying with a privacy laws.
- Can help the organization to grow and get more customers to chose them based on their security and privacy compared to other competitors.

Question 2.9:	Discuss the concept of a shared responsibility model in the context of this data breach incident. Explain the roles and responsibilities of both the cloud service provider and the healthcare service in ensuring the protection of personal information. Write your answer in 200-250 words.

Your response:

When it comes to shared responsibility model in the context of this data incident, both the cloud service provider and the healthcare have their part regarding the protection of personal information. Each party has its own responsibility as the following:

The cloud service provider requires to maintain the infrastructure and their services to ensure their integrity which includes strong encryption, access controls and regular security audits to protect any data on the servers. The service provided also required to regularly assessing vulnerabilities and creating updated incident response plans.

The healthcare service provided should take the right steps and research the best service provider which is complying will all laws and regulations to ensure the security of the data. Also, it is important to review the service provider regularly to ensure the ongoing compliance with laws and agreements. Furthermore, the healthcare service needs to ensure that they are not storing unnecessary data about individual which might not be necessary for operations. The data stored needs to be reviewed regularly and removed if not necessary. The healthcare service also needs to keep a good communication with customers, clients and stakeholders and inform them with any incident that occurred.

Resources:

https://www.commvault.com/blogs/understanding-the-shared-responsibility-model#:~:text=The%20Shared%20Responsibility%20Model%20(SRM,applications%20within%20the%20cloud%20environment.

Question 2.10:	Discuss the importance of insurance for cloud data and services and why it is recommended to appropriate organizational bodies. What are the potential benefits and considerations of having insurance coverage for cloud-related incidents, such as data breaches? Write your answer in 200-250 words.

Your response: Some incidents could cause severe consequences such as financial and operational. The right insurance could assist the organization to manage those consequences and minimize their impacts. Dealing and mitigating a data breach could be expensive due to legal fees, fines and the compensation of affected customers along with mitigation cost which the organization might not be able to recover from. The insurance could help the organization to recover from the incidents and minimize the loss. Also, having insurance encouraging the organization to have risk management in place and best practices in regard of data security as it's a requirement before insuring the organization. Furthermore, the insurance company can help organization to repair their reputational damages and maintain the trust of the customers.

While having an insurance can be beneficial, it is important to take into consideration the fact that every insurance has coverage limits depending on the cover the organization choose so it is very important to make sure the insurance is covering any potential risks by researching the right policy. When choosing an insurance company, it is critical to understand all conditions and policies and check if they are covering with the potential risks of the organization.

Resources:
https://www.majesco.com/blog/five-benefits-of-cloud-computing-for-insurers/

Question 2.11:	Perform a web search and identify the Australian legal requirement for the Cloud Service provider to protect the customer data?

Your response:
The legal requirements are:

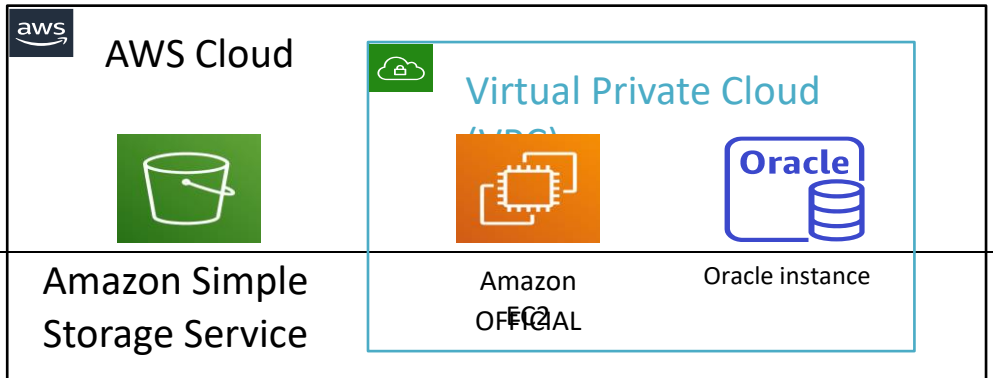
- Privacy Act 1988: This is the main legal requirement for data protection. It gives the organization a guidance of steps which needs to be followed to protect data. It is required to follow all Australian Privacy Principles to ensure the protection of the data.
- Security of Critical Infrastructure Act 2018: Outlining the obligations of organization when managing critical infrastructure to make sure they can response to cybersecurity incidents and mitigate them quickly.
- Telecommunication Act 1979: To ensure the security of telecommunication services when data is transmitted over cloud services.

Resources:
https://iclg.com/practice-areas/cybersecurity-laws-and-regulations/australia

Part-3 Create, Assess and Secure Cloud Services.
To complete this section, you will need to utilize an **AWS Academy account**.
URL for login to AWS Academy: https://awsacademy.instructure.com/login/canvas
If you don't have an account, please request assistance from your teacher.

When responding to the questions, please make sure to include relevant screenshots of your work alongside your explanations. This will help improve clarity and understanding of your answers.

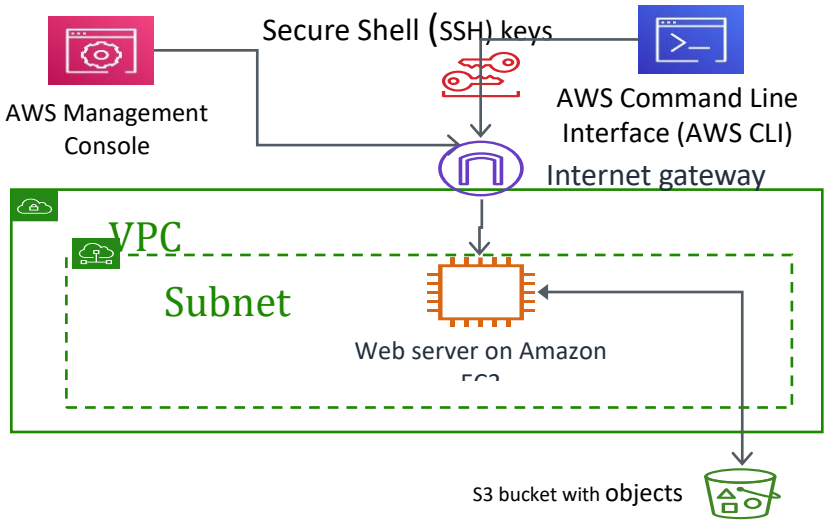
Q3.1 Consider the case where a customer uses the AWS services and resources that are shown here. Who is responsible for maintaining security? AWS or the customer?



AWS Global Infrastructure

QNO.	Question	Response
1	Upgrades and patches to the operating system on the EC2 instance?	Your response: Customer
2	Physical security of the data center?	Your response: AWS
3	Virtualization infrastructure?	Your response: AWS
4	EC2 security group settings?	Your response: Customer
5	Configuration of applications that run on the EC2 instance?	Your response: Customer
6	Oracle upgrades or patches If the Oracle instance runs as an Amazon RDS instance?	Your response: AWS
7	Oracle upgrades or patches If Oracle runs on an EC2 instance?	Your response: Customer
8	S3 bucket access configuration?	Your response: Customer

Q3.2 Consider the case where a customer uses the AWS services and resources that are shown here. Who is responsible for maintaining security? AWS or the customer?



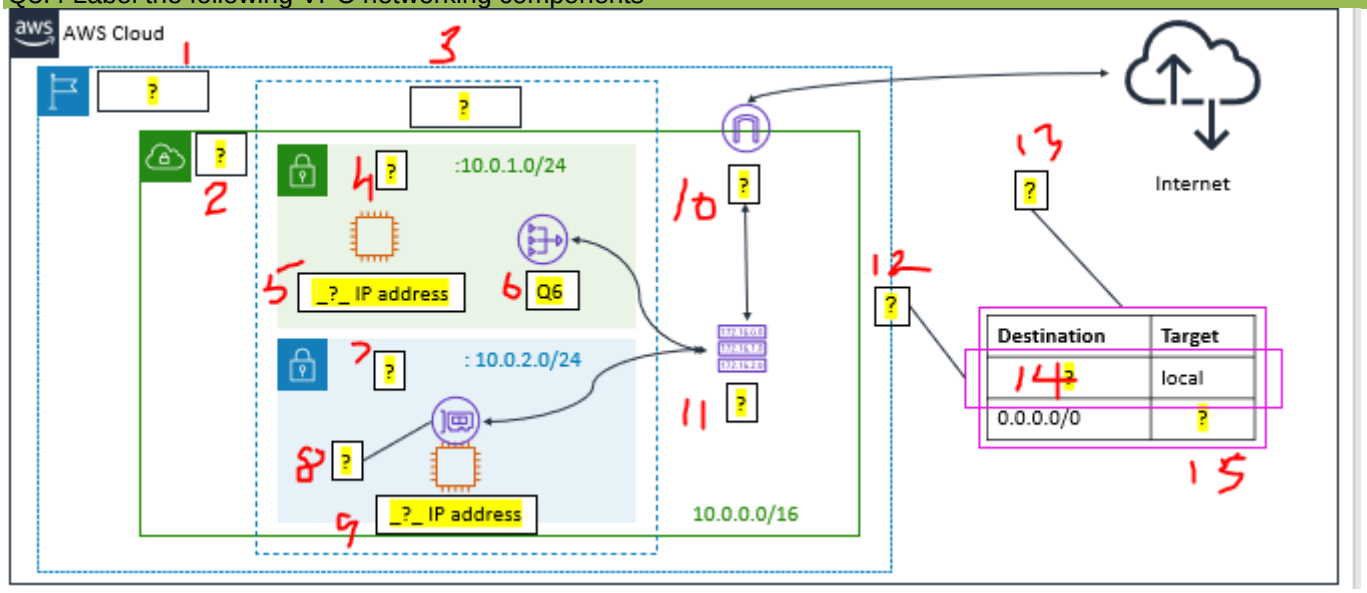
QNO.	Question	Response
1	Ensuring that the AWS Management Console is not hacked?	Your response: AWS
2	Configuring the subnet?	Your response: Customer
3	Configuring the VPC?	Your response: Customer
4	Protecting against network outages in AWS Regions?	Your response: AWS
5	Securing the SSH keys	Your response: Customer
6	Ensuring network isolation between AWS customers' data?	Your response: AWS

7	Ensuring low-latency network connection between the web server and the S3 bucket?	Your response: AWS
8	Enforcing multi-factor authentication for all user logins?	Your response: Customer

Q3.3 Research on IAM service and provide description of each component listed below.

IAM Component	***Description***
IAM User	Your response: IAM user is a component which allows individuals or systems to access to the AWS management console, CLI or APIs to manage the service. Every user got its unique credential and every user associated with a specific permission which were given depending on the resources they are allowed to access or manage.
IAM group	Your response: It is a collection of IAM users which are managed together as a group and assigned with the permissions which were given to them as a group. The group can access only the permissions they were given. It helps administrators to manage users' permission in an easy way from one place instead of giving permission to each user by itself.
IAM policy	Your response: Helps administrators to assign policies and define permissions for IAM user, group or role. Those policies can define who is allowed or denied accessing certain policies within the AWS management console.
IAM role	Your response: It is a short-term access control which is not required to have a permanent credentials. It allows administrators to give permissions to anyone who needs to have access to the resources temporarily. IAM role can define which resources are allowed to be accessed with those temporary credentials as well.

Q3.4 Label the following VPC networking components

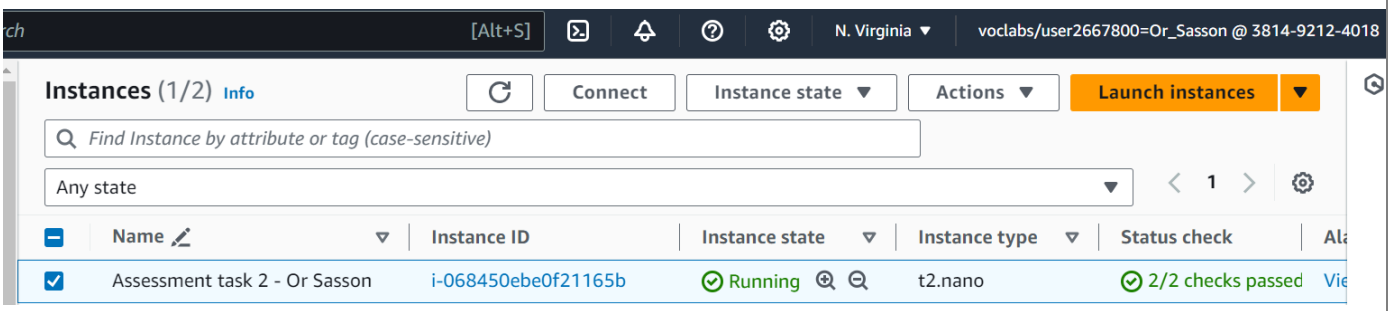


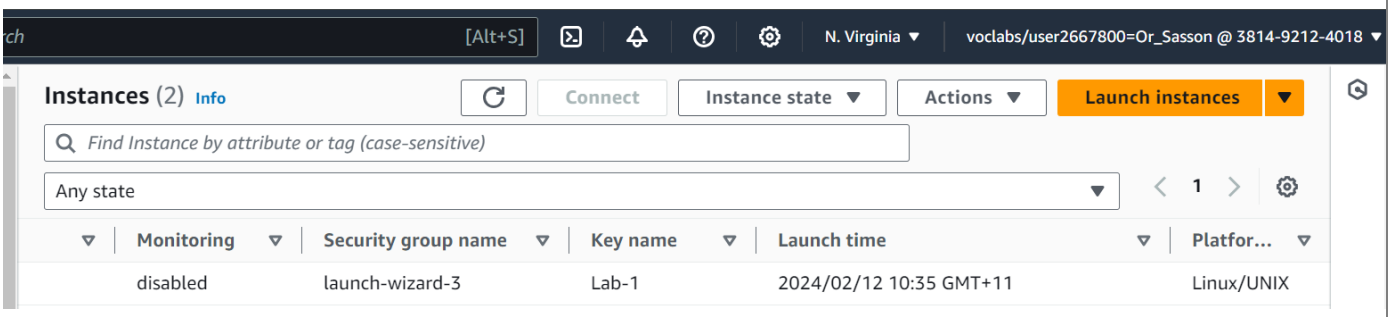
Label No	**Description**
1.	Your response: Region
2.	Your response: VPC
3.	Your response: Availability Zone

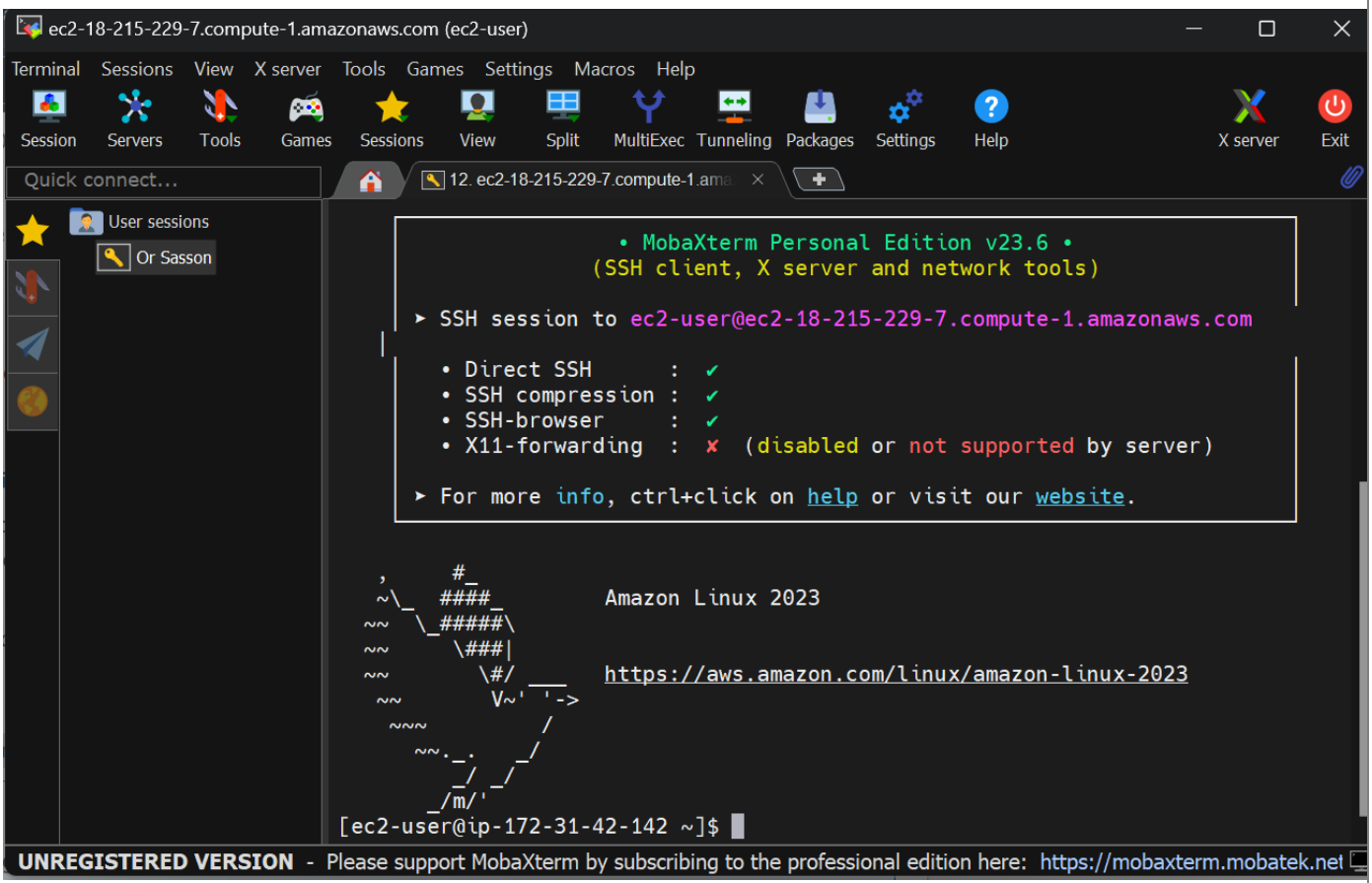
4.	Your response: Public subnet
5.	Your response: Private IP address
6.	Your response: NAT gateway
7.	Your response: Private subnet
8.	Your response: Elastic network interface
9.	Your response: Private IP address
10.	Your response: Internet gateway
11.	Your response: Route table
12.	Your response: Route
13.	Your response: Route Table
14.	Your response: 10.0.0.0/16
15.	Your response: igw-id

Q3.5 Create and launch Linux based EC2 instance in AWS and answer following questions.

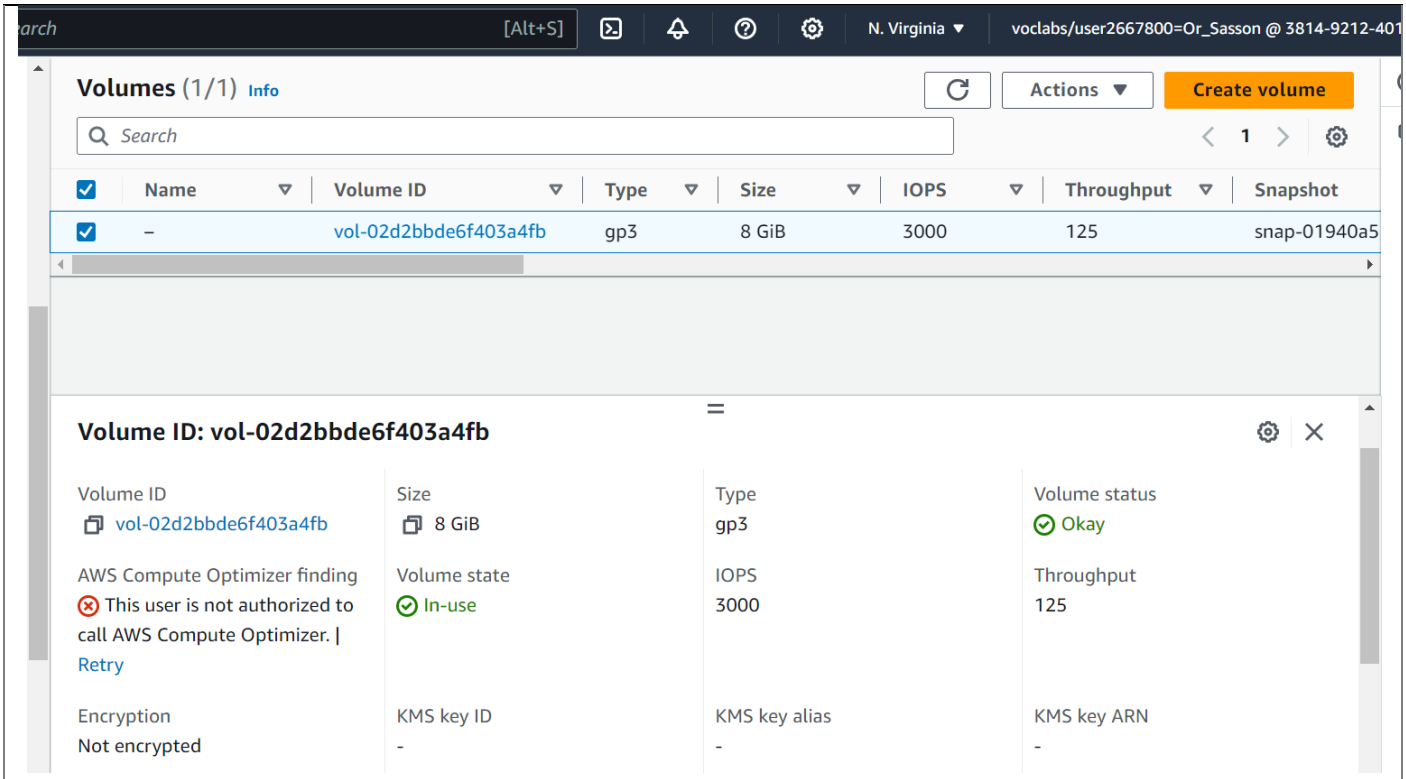
Screenshot of the EC2 console showing your running VM

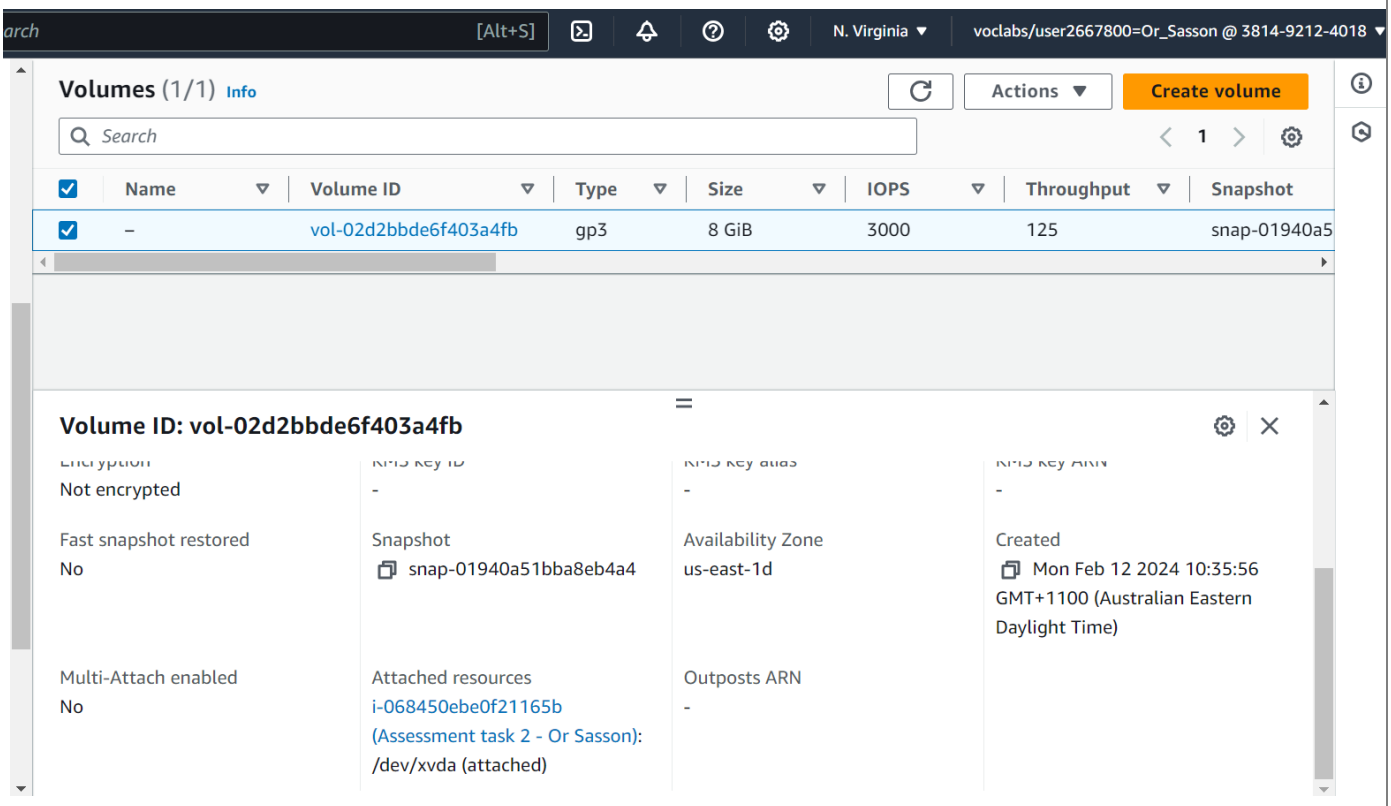




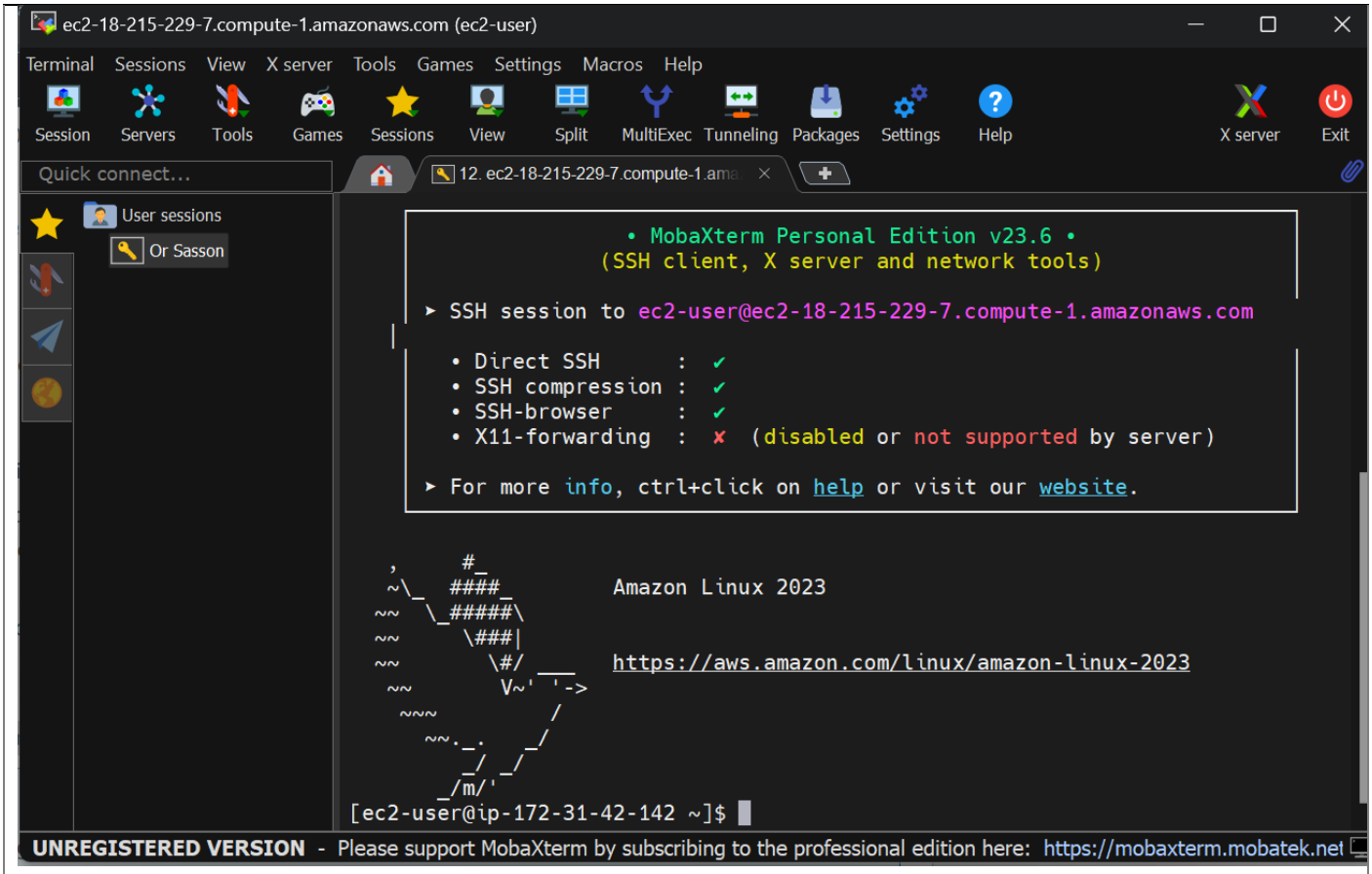


Screenshot of the EC2 console showing the EBS volume that is the disk image for your VM (Go to Elastic Block Store -> Volumes)



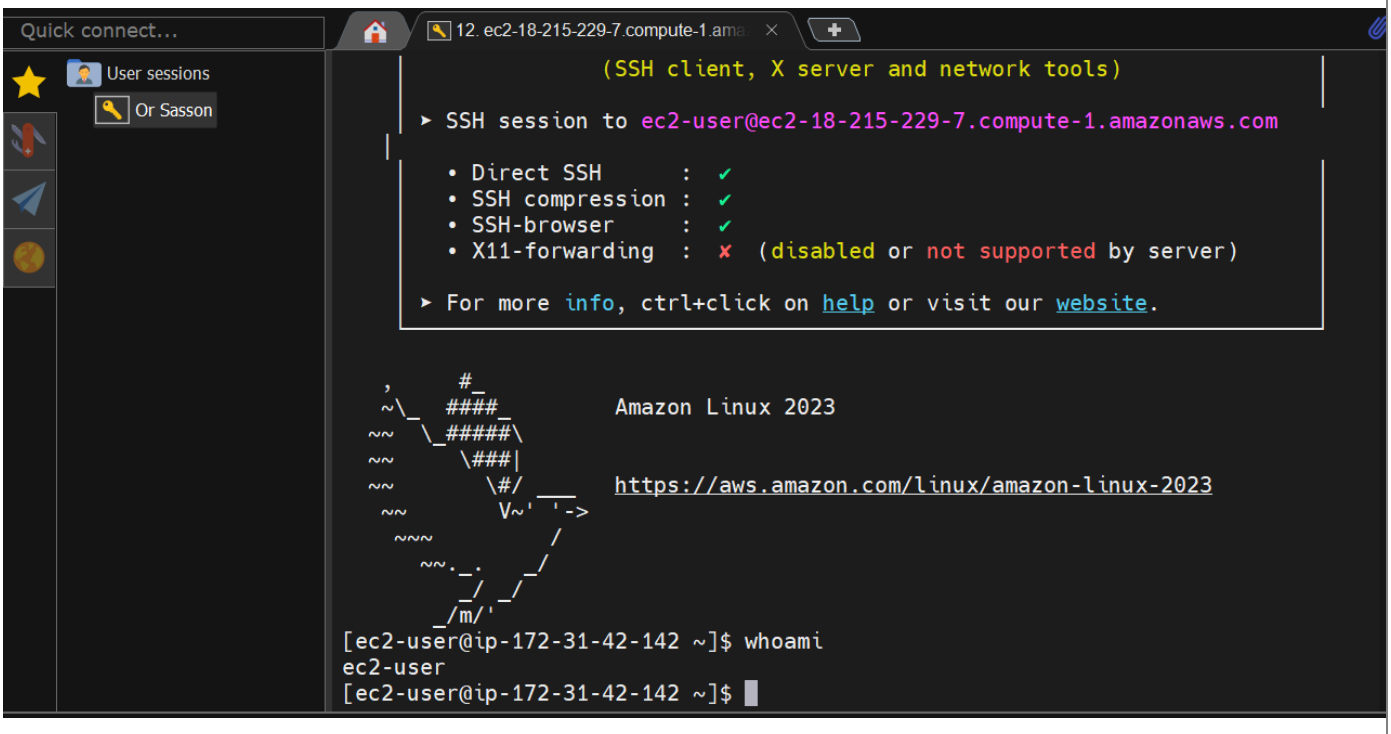


Screenshot of the SSH console showing you logged into your VM and at the command prompt

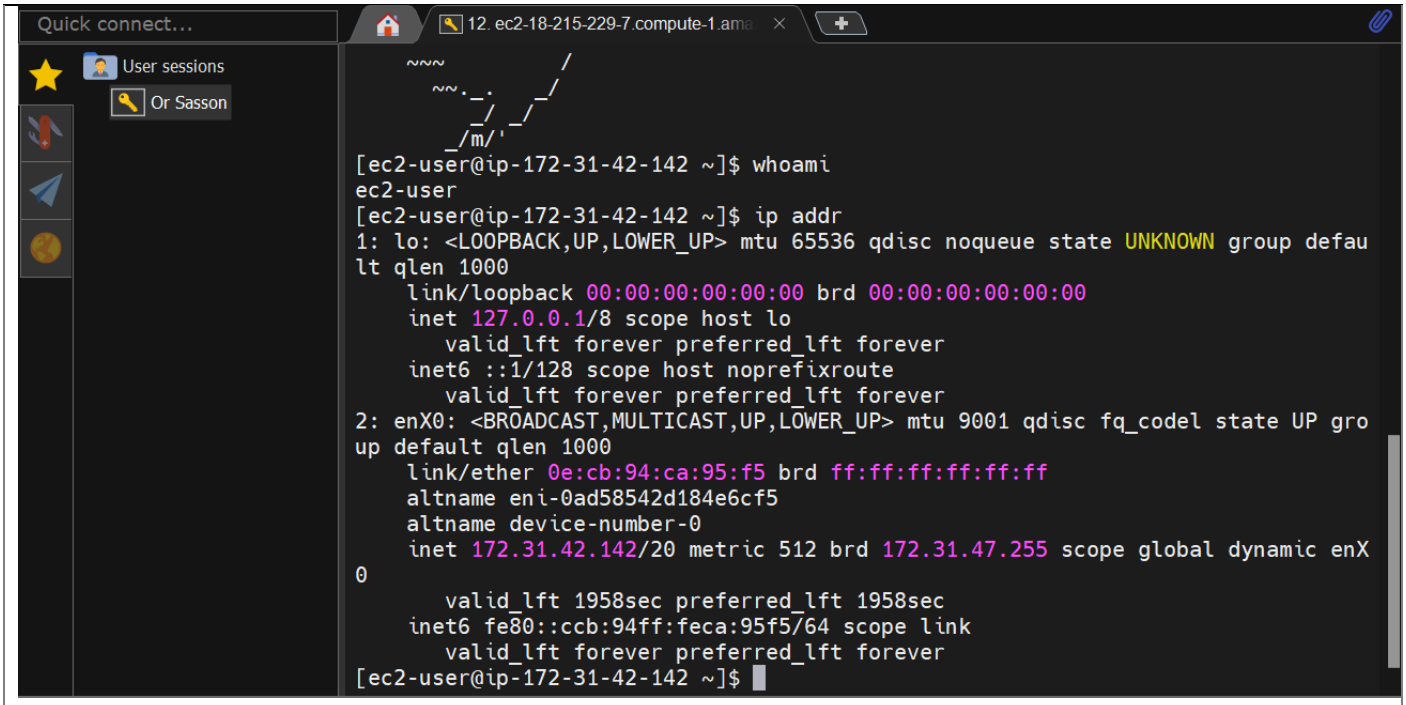


#Screenshot of the SSH console showing the following information (enter all these commands in your SSH terminal)

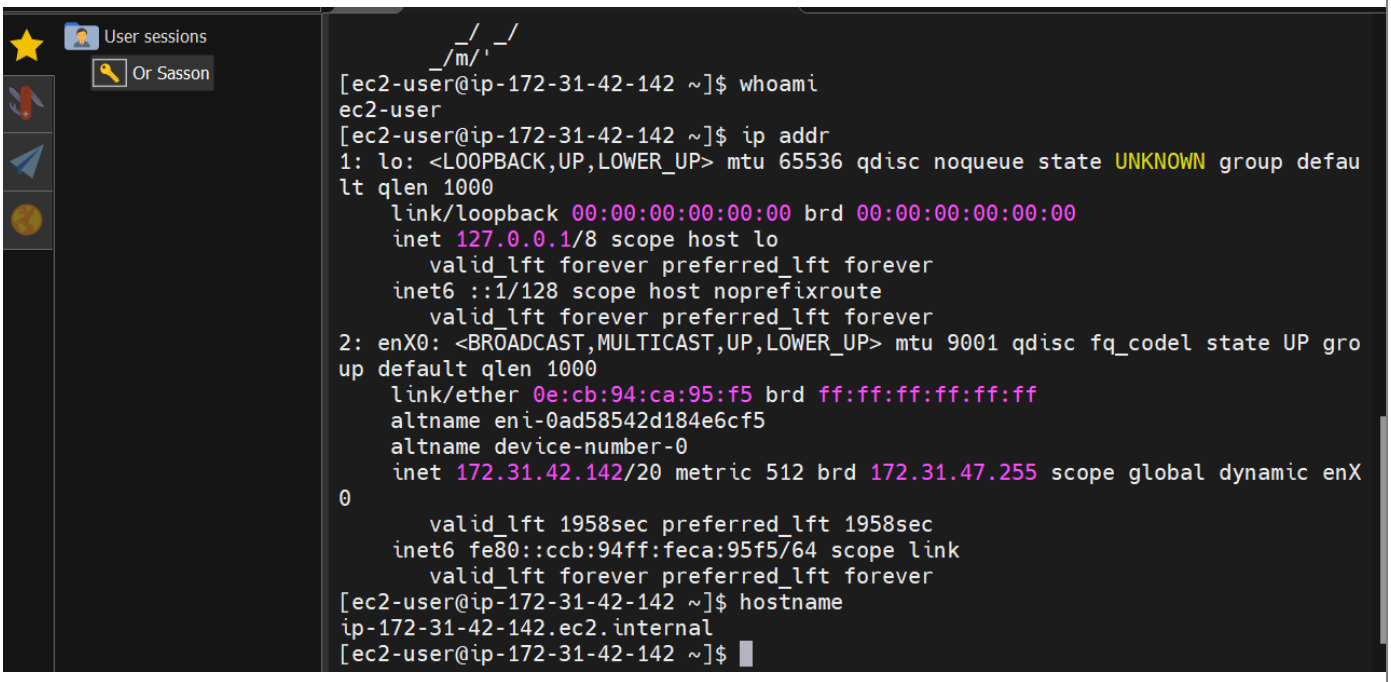
- *What is my username? Whoami*



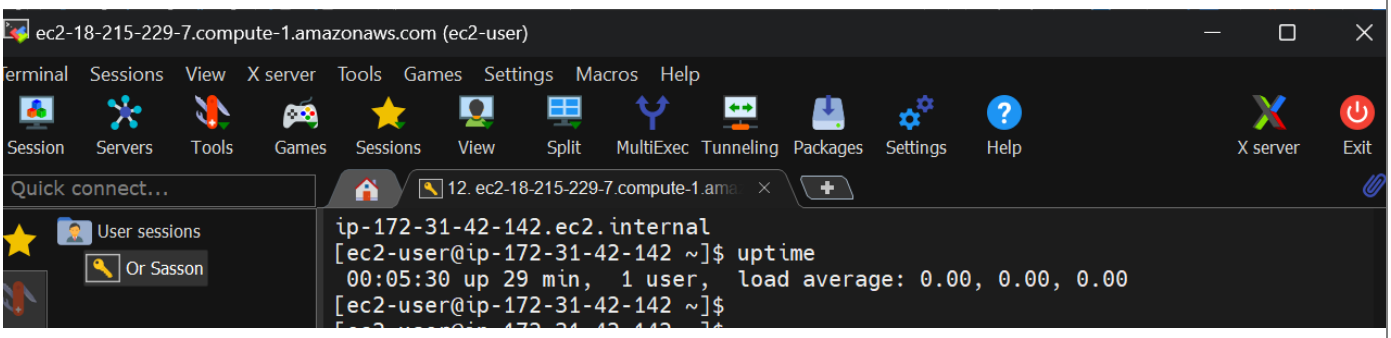
- *What is my private (AWS internal) IP address? ip addr*



- *What is the hostname of my VM? Hostname*



- *How long has the system been running? Uptime*



Q3.6 Install an NGINX Web Server and Load Sample website on the webserver. (You can find sample website on Brightspace or ask your assessor to provide you the sample website).

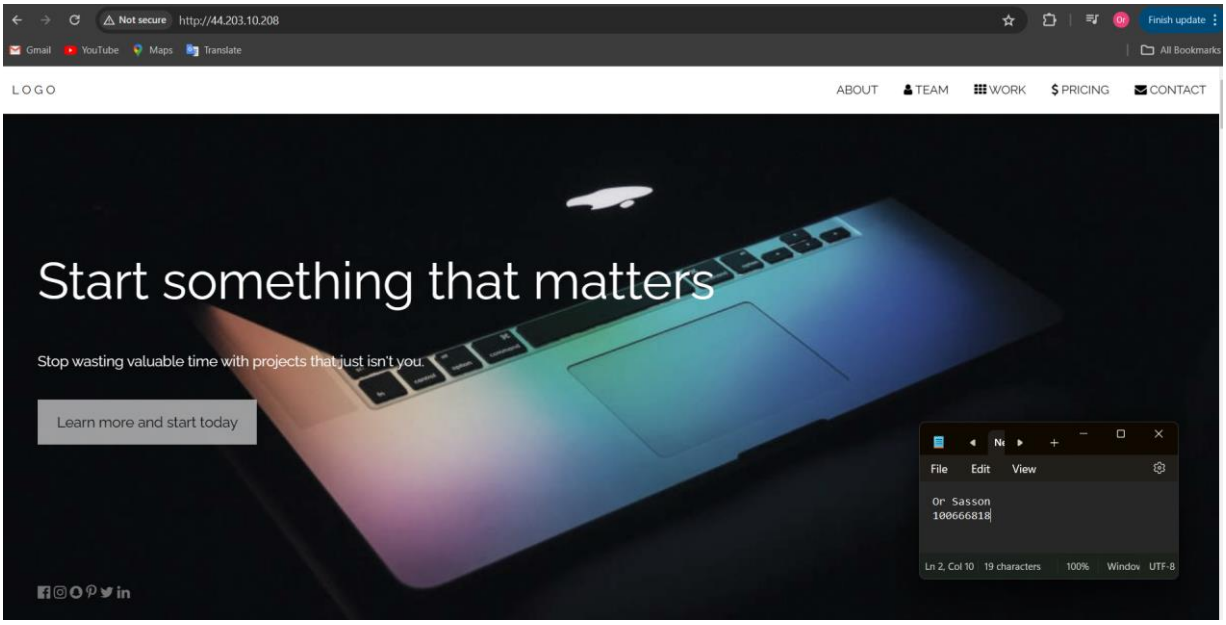
#Which version of Nginx was installed from the Ubuntu package manager?

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-10-101-14-42:~$ sudo nginx -v
nginx version: nginx/1.24.0 (Ubuntu)
ubuntu@ip-10-101-14-42:~$
```

i-0c48debf49767f48a (Web Server- Ubuntu - Or Sasson)

PublicIPs: 44.203.10.208 PrivateIPs: 10.101.14.42

Screenshot of the sample website successfully viewed in your web browser



Q3.7 To access Sample website on NGINX Web Server in Q3.6, What Inbound rules you have configured to access the website.

screenshot of the security group

I configured the inbound rule HTTP port 80 and accessed the website through this port.

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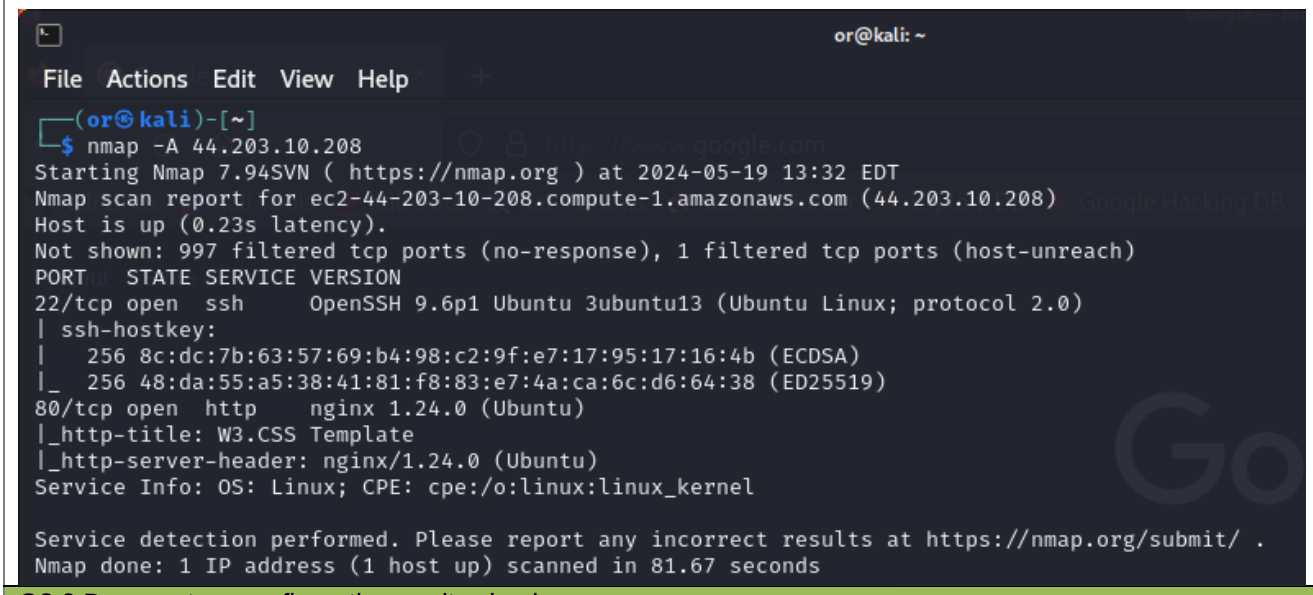
i-0c48debf49767f48a (Web Server- Ubuntu - Or Sasson)

Filter rules

Security group rule ID	Port range	Protocol	Source	Security groups
sgr-02d6eedc2b75692d3	22	TCP	0.0.0.0/0	Tiger - SG - Webserver
sgr-06a02d907261078c5	0	TCP	0.0.0.0/0	Tiger - SG - Webserver
sgr-045d55854579d51fa	80	TCP	0.0.0.0/0	Tiger - SG - Webserver
sgr-0774cc7becb1422c8	443	TCP	0.0.0.0/0	Tiger - SG - Webserver

Q3.8 Run a security scan of the web server

Provide the text output of the Nmap security scan of your web server in "ALL" mode (which runs OS and Service Detection)- you can use Kali linux to scan webserver



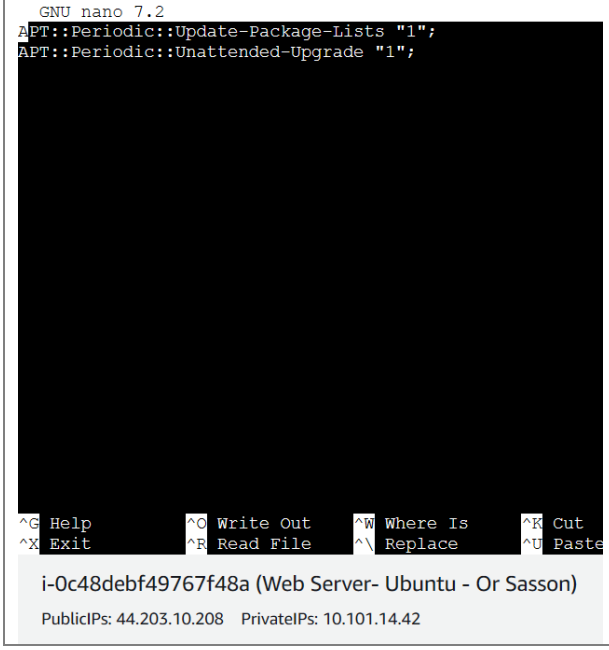
Q3.9 Do a system configuration sanity check

- Confirm that the webserver is set to start automatically at bootup
- Confirm that OS is set to automatically apply security updates

Provide screenshot(s) documenting that the web server is set to start automatically at system boot, and that the OS is set to automatically apply security updates on a regular schedule.



As shown in the screenshot below, there is the number 1 next to the updates which means that automatic updates are enabled.



Q3.10 What are the tradeoffs between having a server OS automatically apply security updates versus having an administrator manually install updates?

Maximum 100 words

When you enable automatic security updates, the security updates will be applied immediately after they released which means the system will be updated against new vulnerabilities which were covered on the recent report. Also, the administrators won't need to worry about the updates as they will be applied automatically. Also, the IT staff could focus on other tasks, rather than updating systems. On the other hand, automatic updates could interrupt operations and cause downtime, especially if it requires reboot after the update. Furthermore, the administrator won't be able to test the updates before applying it to make sure there are error free.

Manual security updates giving a better control for administrators. The administrator can test the updates before deploying it, decide when to apply them, custom configuration and make sure they are applied during off-peak hours. Also, they can decide which update is a priority and update accordingly. However, manually updating can put the system in risk sue to delaying the updates and not updating the system as soon as the update is available. Also, there is a chance to face human errors. Malual updates requires effort and time from the IT staff as well.

Q3.11 What's going to happen when our website is too popular for single server to manage?

Maximum 100 words

The website can experience performance issues including crashes in case of high demand on one server as it will be overwhelmed. It can be avoided if we apply vertical scaling with more available resources such as CPU, RAM and storage along with Horizontal scaling adding more server to balance the demand. Load balancing could prevent such an event by balancing the load evenly between the available server to make sure the website is not relying on a single server which can get overwhelmed due to high demand.

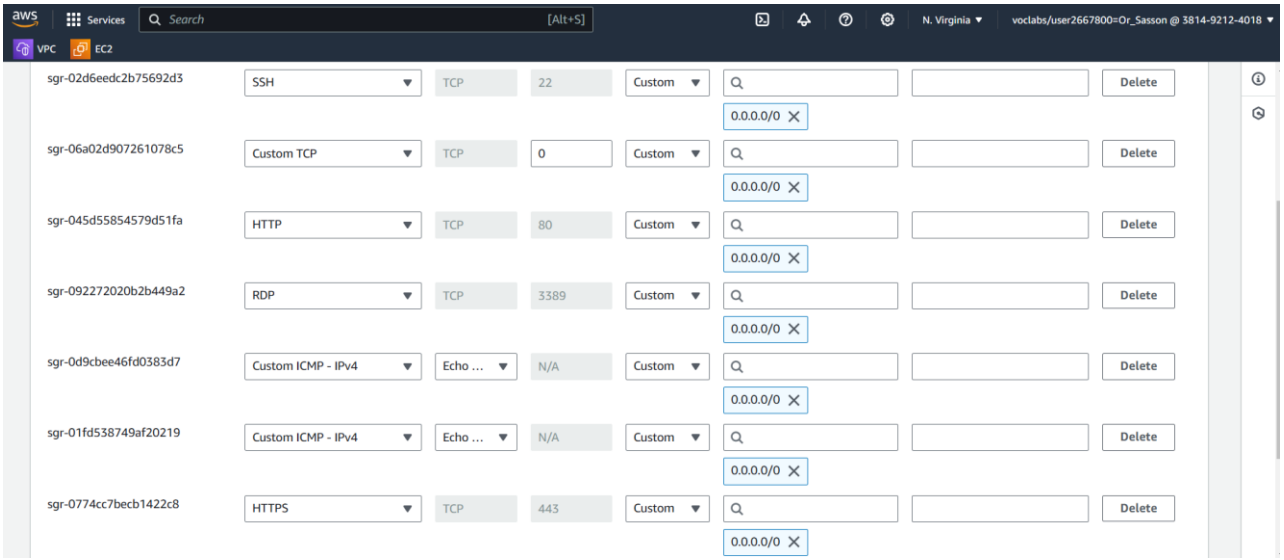
Q3.12 What happens, if our single webserver fails? Is our website suddenly inaccessible?

Maximum 100 words

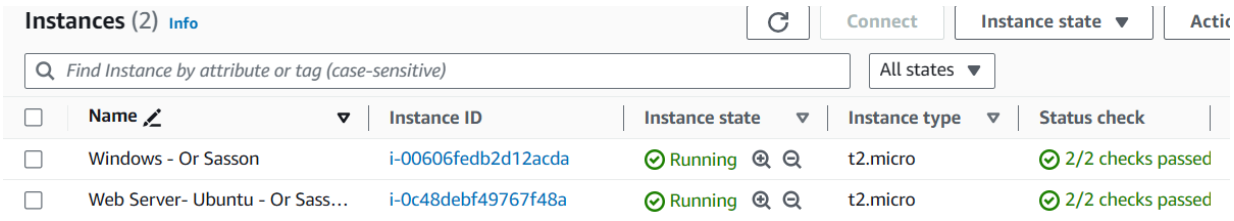
If the webserver fails it can lead to a bad customers experience which can lead to a lose of customers. Also, due to the downtime and customers lost, can lead to a financial damage and damage the reputation of the business. Redundancy and high availability with load balancers along with Horizontal scaling, could minimize the risk for downtime and ensure the accessibility to the website.

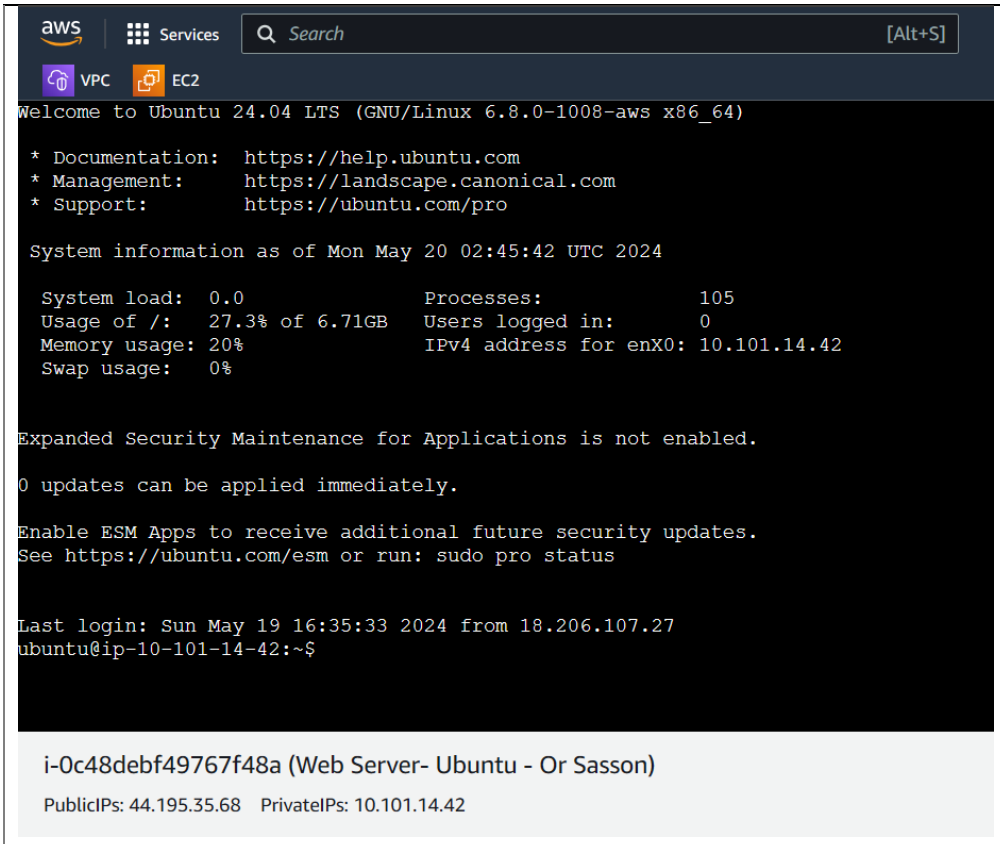
Q3.13 Create two EC2 instances (windows and linux) and ping each other

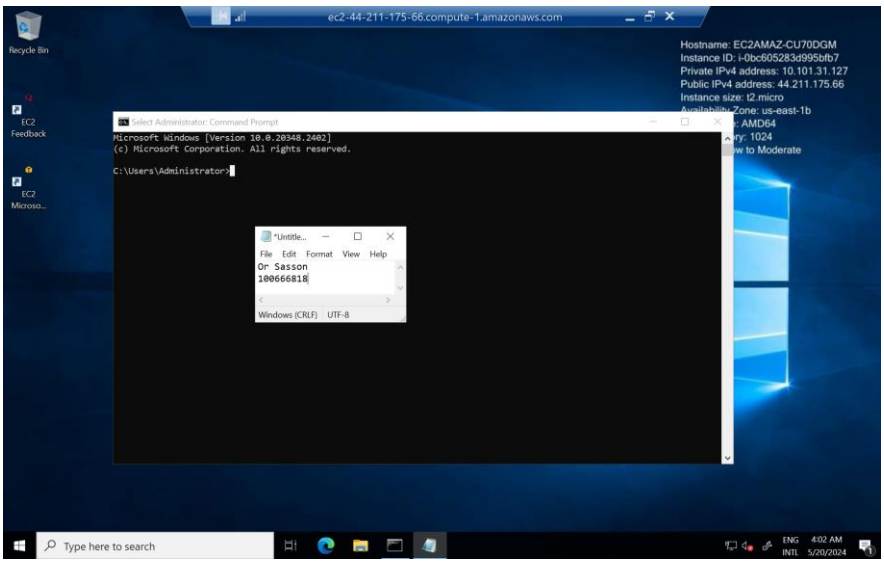
Screenshot of security group.



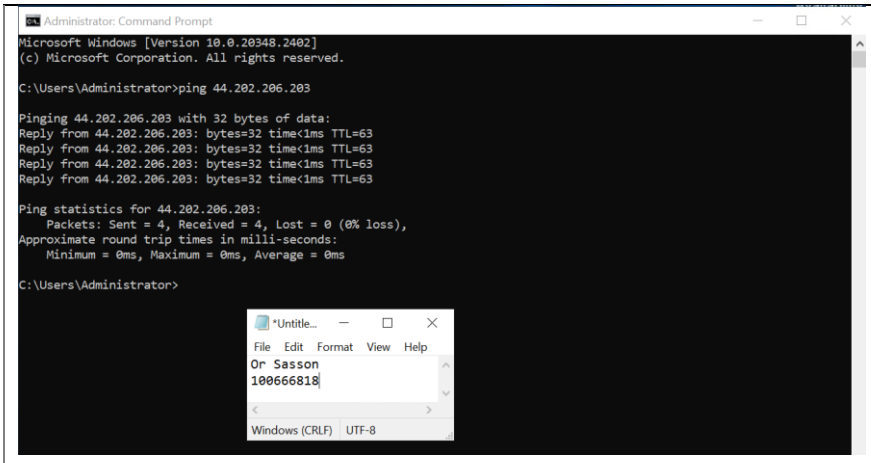
Screenshot of Running Both EC2 instances



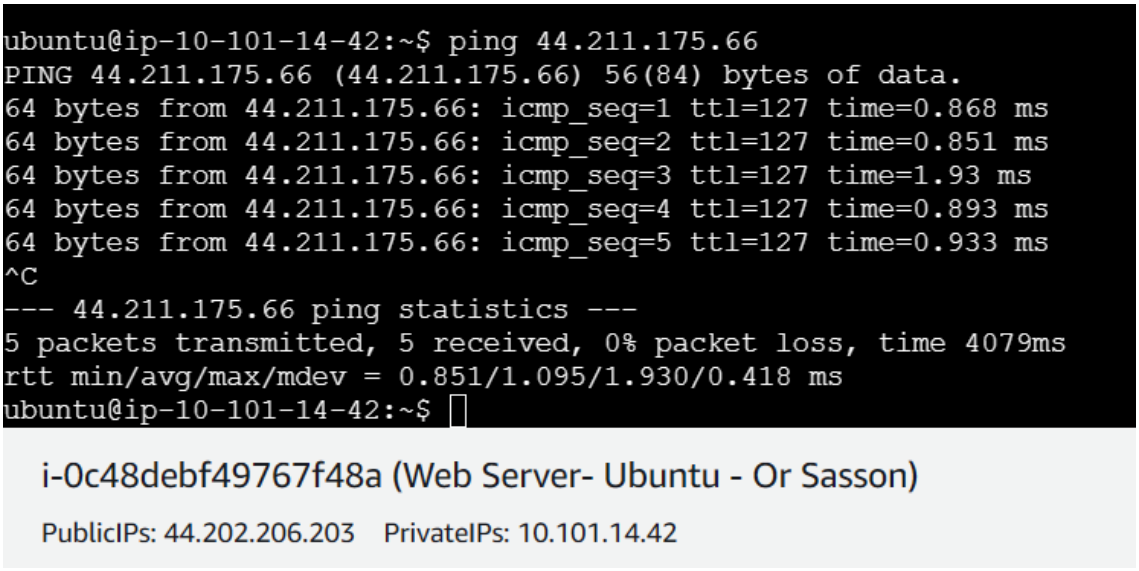




#Screenshot of ping results from windows to Linux



Screenshot of ping results from Linux to windows

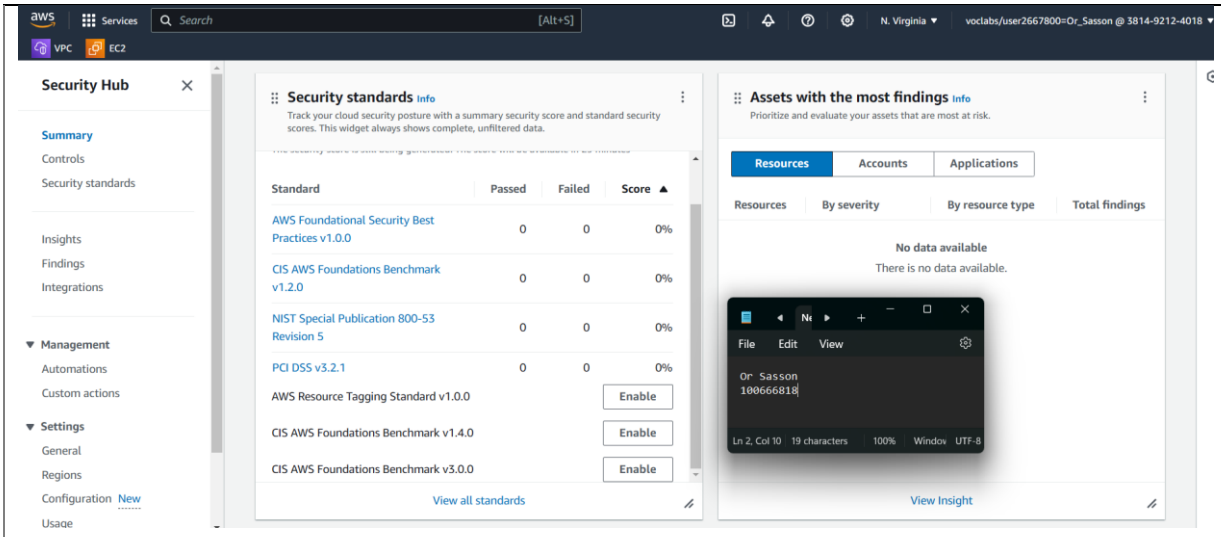


Q3.14 What is Security Hub, how to enable Security Hub.

Maximum 50 words

The Security Hub gives a centralized overview of the security of the AWS and help knowing whether the security state of the instances and giving a good indication about the compliance with security industry standards and best practices. The Security Hub collecting information from a range of AWS services as well as third party. It also can automate responses to security findings.

Screenshot of security hub

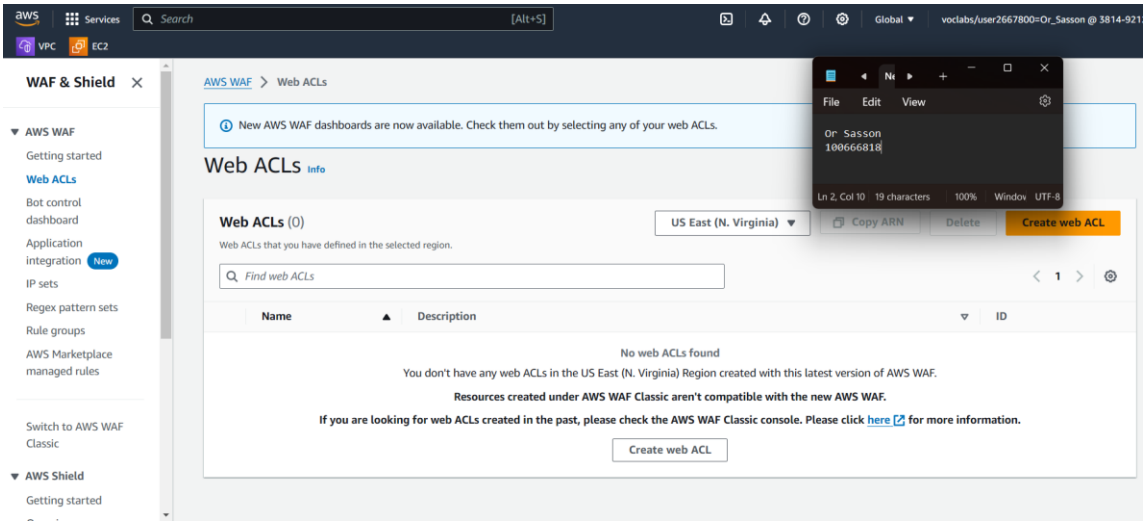


Q3.15 Briefly explain WAF and provide screenshot of AWS WAF dashboard.

Explanation of WAF

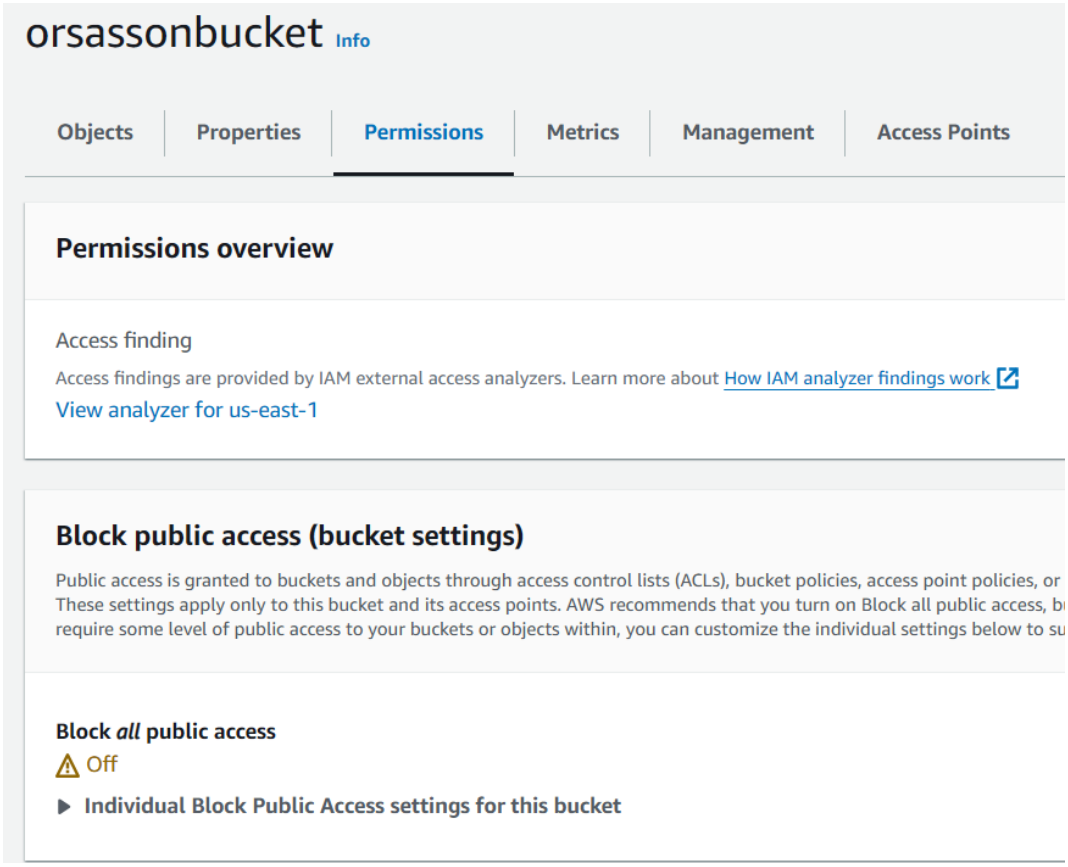
Web Application Firewall giving the ability to create security rules and control incoming web requests by configuring ACLs rules. The WAF is monitoring the traffic in real time and blocking malicious traffic based on the user's rules configurations. It is integrated with other AWS services perfectly for great results and security.

#Screenshot of AWS WAF console

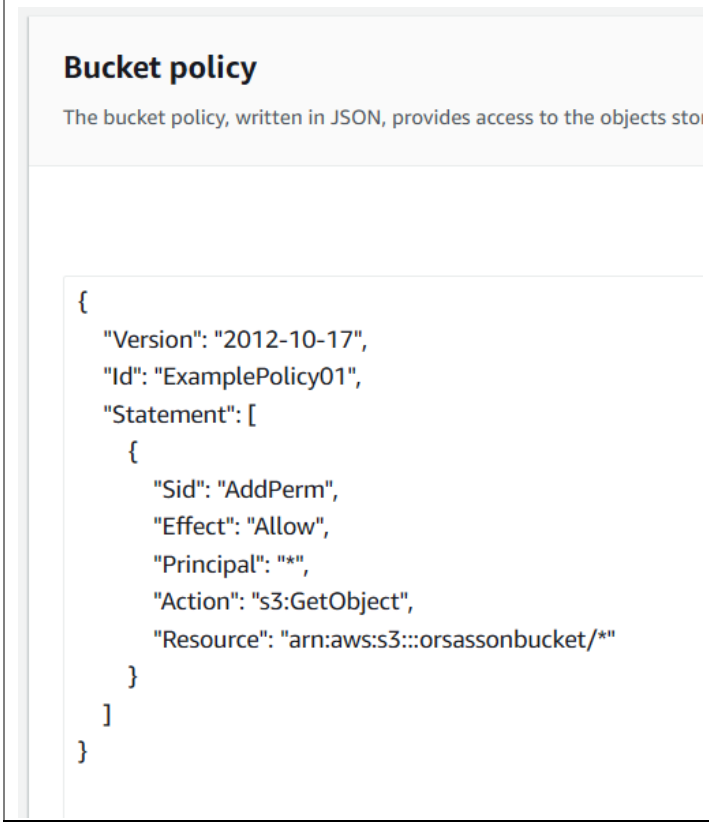


Q3.16 Create S3 Bucket and answer following questions.

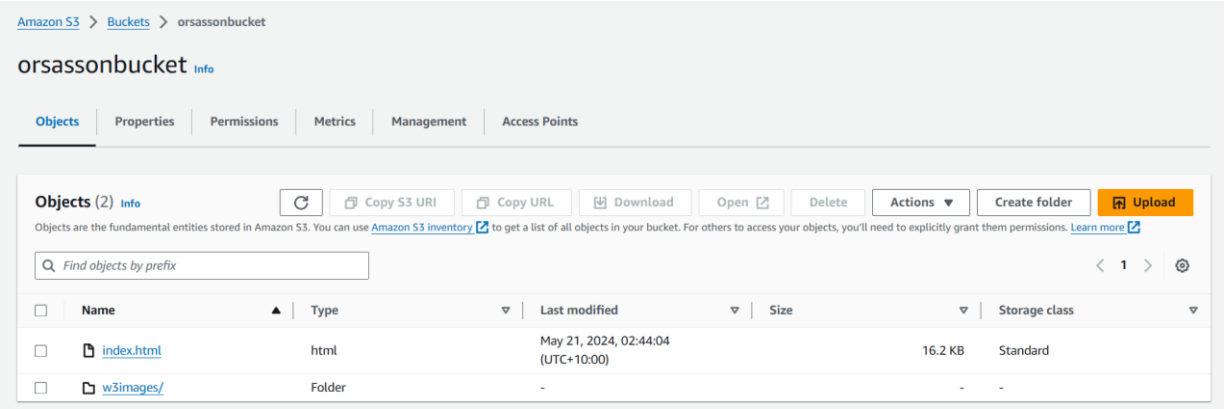
Screenshot of S3 bucket with publicly accessible



Screenshot of bucket policy

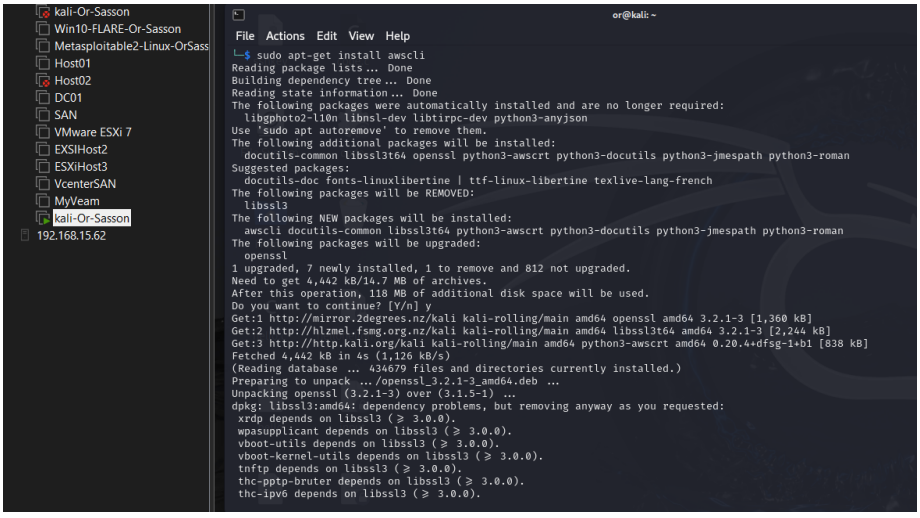


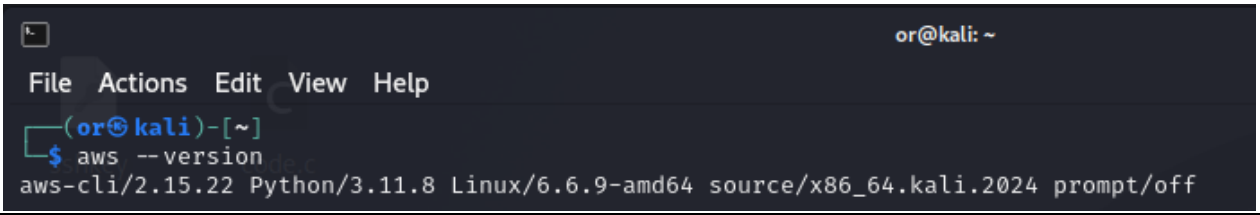
#Screenshot of file uploaded on S3 bucket (you can upload any txt file)



Q3.17 S3 is a simple storage service in AWS, which is used to store data that is frequently accessible. Any misconfiguration to S3 can expose any sensitive data. Go to flaws.cloud capture the flag website and solve level-1 challenge and answer following questions by using aws cli.

screenshot of awscli installation





what is ip address of flaws.cloud

```
┌──(root㉿kali)-[/home/or]
└─# dig flaws.cloud

; <<>> DiG 9.19.21-1-Debian <<>> flaws.cloud
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 42268
;; flags: qr rd ra; QUERY: 1, ANSWER: 8, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0×0005, udp: 512
;; QUESTION SECTION:
;flaws.cloud.                   IN      A

;; ANSWER SECTION:
flaws.cloud.            5       IN      A       52.218.132.243
flaws.cloud.            5       IN      A       52.218.234.50
flaws.cloud.            5       IN      A       52.92.155.123
flaws.cloud.            5       IN      A       52.92.185.67
flaws.cloud.            5       IN      A       52.92.154.27
flaws.cloud.            5       IN      A       52.92.179.219
flaws.cloud.            5       IN      A       52.92.187.171
flaws.cloud.            5       IN      A       52.92.184.107

;; Query time: 252 msec
;; SERVER: 192.168.15.2#53(192.168.15.2) (UDP)
;; WHEN: Mon May 20 13:41:38 EDT 2024
;; MSG SIZE  rcvd: 168
```

Identify DNS record of flaws.cloud (use dnsrecon)

```
┌──(root㉿kali)-[/home/or]
└─# dnsrecon -d flaws.cloud
[*] std: Performing General Enumeration against: flaws.cloud...
[-] DNSSEC is not configured for flaws.cloud
[*]      SOA ns-1890.awsdns-44.co.uk 205.251.199.98
[*]      SOA ns-1890.awsdns-44.co.uk 2600:9000:5307:6200::1
[*]      NS ns-1061.awsdns-04.org 205.251.196.37
[*]      NS ns-1061.awsdns-04.org 2600:9000:5304:2500::1
[*]      NS ns-1890.awsdns-44.co.uk 205.251.199.98
[*]      NS ns-1890.awsdns-44.co.uk 2600:9000:5307:6200::1
[*]      NS ns-448.awsdns-56.com 205.251.193.192
[*]      NS ns-448.awsdns-56.com 2600:9000:5301:c000::1
[*]      NS ns-966.awsdns-56.net 205.251.195.198
[*]      NS ns-966.awsdns-56.net 2600:9000:5303:c600::1
[*]      A flaws.cloud 52.92.248.187
[*]      A flaws.cloud 52.92.195.219
[*]      A flaws.cloud 52.92.236.107
[*]      A flaws.cloud 52.92.139.59
[*]      A flaws.cloud 52.218.232.139
[*]      A flaws.cloud 52.92.238.59
[*]      A flaws.cloud 52.92.224.179
[*]      A flaws.cloud 52.218.246.50
[*] Enumerating SRV Records
[-] No SRV Records Found for flaws.cloud
```

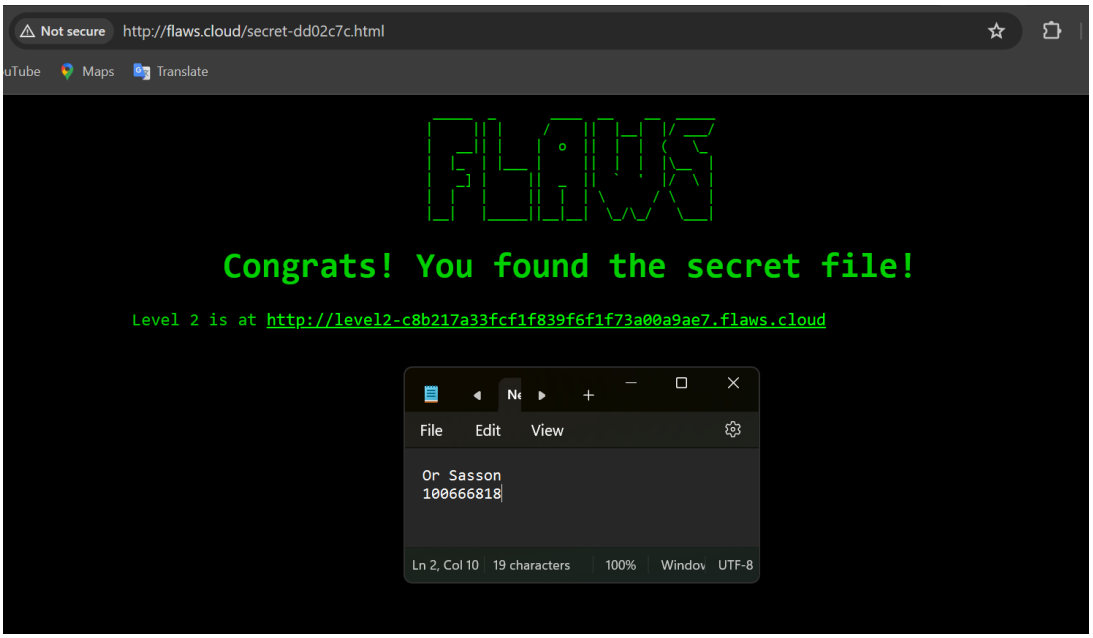
find actual URL of S3 bucket by using dnslookup

```
┌──(root㉿kali)-[/home/or]
└─# nslookup 52.218.205.154
154.205.218.52.in-addr.arpa     name = s3-website-us-west-2.amazonaws.com.
```

#Screenshot of list of all files on s3://flaws.cloud

```
┌──(root㉿kali)-[/home/or]
└─# aws s3 ls  s3://flaws.cloud/ --no-sign-request --region us-west-2
2017-03-13 23:00:38       2575 hint1.html
2017-03-02 23:05:17       1707 hint2.html
2017-03-02 23:05:11       1101 hint3.html
2024-02-21 21:32:41       2861 index.html
2018-07-10 12:47:16      15979 logo.png
2017-02-26 20:59:28         46 robots.txt
2017-02-26 20:59:30       1051 secret-dd02c7c.html
```

#Screenshot of completion of level 1 challenge.


Not secure
http://flaws.cloud/secret-dd02c7c.html
uTube
Maps
Translate
Congrats! You found the secret file!
Level 2 is at http://level2-c8b217a33fcf1f839f6f1f73a00a9ae7.flaws.cloud
File
Edit
View
Or Sasson
100666818
Ln 2, Col 10
19 characters
100%
Windov
UTF-8